



Original software publication

edgar: An R package for the U.S. SEC EDGAR retrieval and parsing of corporate filings

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ABSTRACT

This paper introduces the R package `edgar` to download and analyze the Securities and Exchange Commission's (SEC) mandatory public disclosures in the United States. Corporations in the U.S. submit their periodic reports, registration statements, and financial reports electronically to the SEC. The SEC makes these reports publicly accessible to everyone through the Electronic Data Gathering, Analysis, and Retrieval System (EDGAR). As financial reporting is one of the most crucial aspects of the financial system, efficient retrieval of EDGAR filings becomes imperative for analysts and researchers. We summarize the implementation of `edgar` package that facilitates downloading, parsing, searching, and sentiment analysis of corporate reports.

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Code metadata

Current code version	V2.0.4
Permanent link to code/repository used for this code version	https://github.com/ElsevierSoftwareX/SOFTX_2020_222
Code Ocean compute capsule	none
Legal Code License	GPL
Code versioning system used	git
Software code languages, tools, and services used	R v3.4
Compilation requirements, operating environments & dependencies	R Packages: Rutils, tm, XML, stringr, stringi, qdapRegex
If available Link to developer documentation/manual	R CRAN Manual and Documentation Web link
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1. Motivation and significance

It is noteworthy that the U.S. SEC receives the terabytes of mandatory operation and financial statements, popularly called as filings, every quarter from both public and private firms in the U.S.¹ To download and analyze fundamentals, accounting statements, and future growth possibilities of these firms efficiently are vital. Thus, the `edgar` package [10] developed in R statistical programming software allows researchers, practitioners, and investors to access valuable information from the SEC filings on a large scale.

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¹ The SEC requires a firm to file a particular form type for a specific purpose, listed on <https://www.sec.gov/forms>.

The SEC's public repository known as the EDGAR system, started in 1993, maintains filings for an individual company, mutual fund, and exchange-traded fund (ETF). This platform allows public use of these filings for research, investment, and analysis purposes. However, the EDGAR web interface allows accessing only one filing at a time. To make systematic decisions, `edgar` package helps researchers and analysts to retrieve and parse the required information from these filings in bulk, and performs sentiment analyses.

The SEC has improved its server security, including a significant change in its web interface in 2017. Most of the previously developed packages lack interface to the upgraded EDGAR repository. For example, the links to EDGAR server directories mentioned in [8,9] no longer work. Additionally, these packages do not provide access to multiple files in a single query. For example, `xbrlDoAll` function from XBRL R package only works on

Table 1

Comparative analysis of platforms available for EDGAR mining.

Tool	Platform	Specifications	Limitations
edgarWebR [1]	R	Provides an interface to access the SEC's EDGAR system	-Lacks bulk mining functionality -Only provides the metadata and company information -Lacks parsing of filing for important information -Lacks local storage management
OpenEDGAR [2]	Python	Downloads filings and offers a filing search tool	-Lacks functionality for parsing of important information from filings, such as filing header, 8-K events, business description -Uses outside relational database -Needs computing knowledge to install and use
corpcrawl [3]	Python	Extracts company and its subsidiaries names from 10-K forms	-No storage structure -Package is in beta stage -No help document
python-edgar [4]	Python	Downloads daily index file	-Provides Minimal functionalities -Restriction on form types -Lacks proper error handling
SECEdgar [5]	Python	Downloads 10-K, 10-Q, 8-K, and 13-F forms	
finreportr [6]	R	-Provides filing information of a company -Extracts financial reports from XBRL annual reports	
XBRL [7]	R	Extracts financial statements from XBRL	
[8]	SAS	Downloads filings and provides keyword search function	-Uses licensed software -Restriction on form type -No storage structure
[9]	Perl	-Downloads quarterly indexes -Downloads and analyses 8-K filings	-Requires large operating time -Redundancy in script -Inefficient storage structure -Restrictions on filing types

a single file. Table 1 reports major limitations of the similar tools that provide functionalities to download and parse EDGAR filings. The open-source edgar R package mitigates these limitations and adds new routines. Specifically, it provides an open source tool, access to all filings, robust error handling, better file management system, and scraping and parsing functions. The recent developments in the advancement of research methodologies led to an increase in demand for the usage of such specialized tools.

2. Software description

The edgar package utilizes functions from R.utils [11], tm [12], XML [13], stringr [14], stringi [15], and qdapRegex [16] R packages. A user can install it in R using the following code.

```
install.packages(edgar)
```

2.1. Software functionalities

Table 2 reports all the functions provided by edgar package.

2.2. Software architecture

Efficient download and analysis of a large number of filings require proper storage management. edgar package uses a working directory on a user's machine to store data in a hierarchy structure. It automatically creates all the sub-directories in the selected working directory upon respective function calls. As seen in Fig. 1, other functions in edgar R package call getMasterIndex and getFilings functions to retrieve filing information and download filings from the SEC server, respectively. We recommend, though not mandatory, to maintain the same working directory for every interaction with this package to utilize the existing data.

edgar package stores filing information, complete filings, and extracted data in separate sub-directories illustrated as follows.

- **Daily Indexes:** This directory is generated upon calling the getDailyMaster function and contains daily filing information, also known as daily master index files, in Rda format with filename as *daily_idx_[index date].Rda*.²
- **Master Indexes:** This directory is created upon calling the getMasterIndex function and saves yearly master index in Rda format with filename as *[Year]master.Rda*, e.g., *1994master.Rda*.
- **Edgar filings_full text:** This directory is generated when a user calls the getFilings function. It stores complete filings in text format in separate sub-directories of form types and firm CIK number with filename as *[CIK]_[form type]_[date filed]_[Accession Number]*.³ For example, 10-K statement of a firm with CIK = 38079 can be found at the location "Edgar filings_full text->Form 10-K->38079->38079_10-K_2005-03-15_0001047469-05-006546.txt".
- **Edgar filings_HTML view:** This directory is created upon a call of the getFilingsHTML function and saves filings in HTML format with the filename in format *[CIK]_[form type]_[date filed]_[Accession Number]*. These HTML files are stored in separate sub-directories of form types and firm CIK number. For example, HTML view of 10-K statement in the previous example can be found on filepath "Edgar filings_HTML view->Form 10-K->38079->38079_10-K_2005-03-15_0001047469-05-006546.html".
- **Keyword search results:** This directory is created upon use of the searchFilings function and saves the extracted filing search results in HTML format. The HTML view of search results provides the extracted filing text surrounded by the user keywords. These results for each filing is stored in the format of *[CIK]_[form type]_[date filed]_[Accession Number]*.

² Rda is a native R data structure to store objects of vectors, matrices, and dataframes.

³ The central index key (CIK) acts as a primary identifier for corporations and individuals who file disclosures with the US SEC.

Table 2
Functions provided by *edgar* package.

<code>getDailyMaster</code>	Retrieves daily master index
<code>getMasterIndex</code>	Retrieves quarterly master index
<code>getFilingInfo</code>	Retrieves filing information of a firm
<code>getFilings</code>	Retrieves EDGAR filings from SEC server
<code>getFilingsHTML</code>	Get HTML view of EDGAR filings
<code>getFilingHeader</code>	Scrapes EDGAR filing header information
<code>searchFilings</code>	Searches EDGAR filings for specific words
<code>getBusinDescr</code>	Retrieves business descriptions from annual reports
<code>getMgmtDisc</code>	Retrieves MD&A section from annual reports
<code>get8KItems</code>	Retrieves Form 8-K event information
<code>getSentiment</code>	Provides sentiment measures of EDGAR filings

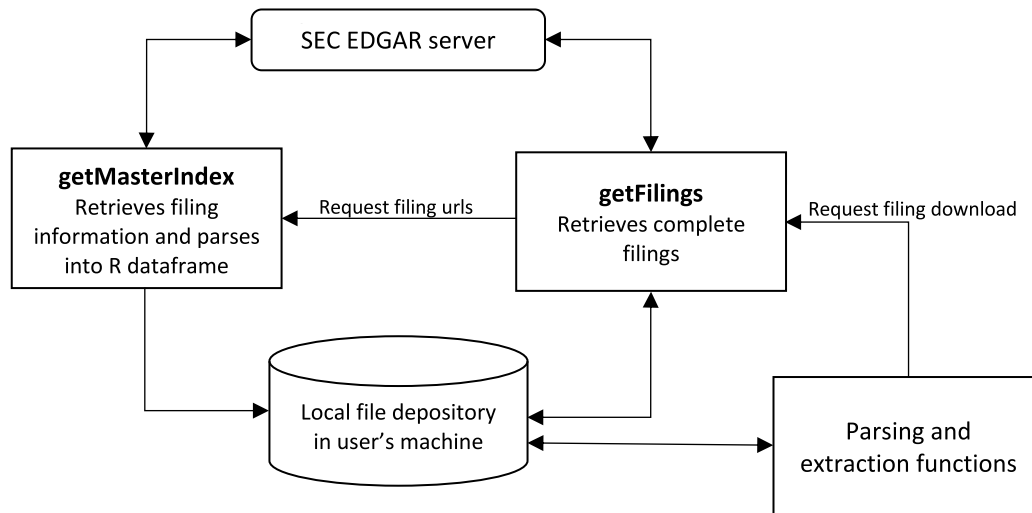


Fig. 1. Architecture of *edgar* R package.

- *Business descriptions text*: This directory stores extracted business description section in text format with filename as `[CIK]_[form type]_[date filed]_[Accession Number]`. It is created with usage of the `getBusinDescr` function.
- *MD&A section text*: This directory saves extracted Management's Discussion and Analysis (MD&A) section in text format with filename as `[CIK]_[form type]_[date filed]_[Accession Number]`. It is created upon a call of the `getMgmtDisc` function.

3. Implementation of *edgar* package

3.1. SEC guidelines on downloading EDGAR files

To avoid load congestion on EDGAR server, the SEC suggests downloading only the required filings. The SEC makes it mandatory for a user to declare user-agent id in the request header. The given link explains this requirement in detail: <https://www.sec.gov/os/accessing-edgar-data>. Accordingly, *edgar* package requires a user to pass user-agent id for all its functions. Thus, a user needs to declare it in the following form:

```
R> useragent = "Your Name Contact@domain.com"
```

3.2. Download filing information from the SEC EDGAR server

The U.S. SEC receives financial reports regularly from various public and institutional firms. The SEC generates quarterly index

files (also called as master index) with the information on all the filings filed on the SEC in a given quarter. The quarterly master indexes are uploaded on the SEC server in `idx` (index) compressed formats on www.sec.gov/Archives/edgar/full-index/, which includes the Central Index Key (CIK) number, company name, form type, date filed, and weblink for financial reports. The `getMasterIndex` function of *edgar* package downloads these quarterly master indexes by taking a vector of years as a user input. This function downloads quarterly master index files, cleans them, consolidates quarterly indexes to yearly, and stores them as yearly master index files in *Rda* format in the directory "*Master Indexes*". A user needs to maintain the same working directory while using *edgar* packages as it utilizes these yearly master indexes to search for filing information and download filings from the EDGAR server. The following code illustrates a use of this function.

```
R> getMasterIndex(2006, useragent)
Downloading Master Indexes from SEC server for 2006 ...
Master Index for quarter 1
Master Index for quarter 2
Master Index for quarter 3
Master Index for quarter 4
```

Similar to the quarterly index files, the SEC also maintains index files for filings filed on the SEC in a given day. These daily information on filing can be obtained using `getDailyMaster` function. Thus function takes a date as an input from a user, downloads and cleans the daily index file, and returns information on daily filings in a dataframe.

3.3. Search for filing information and download filings

The `getFilingInfo` function provides filing information of a firm based on a firm identifier. It takes a desired firm identifier in the form of full/partial firm name or CIK number, filing year(s), filing quarter(s), and form type(s) as input parameters.⁴ The following code demonstrates the usage of this function.

```
R> info <- getFilingInfo('United Technologies',
+   filing.year=c(2005, 2006), quarter=c(1,2),
+   form.type=c('10-K','DEF 14A'), useragent)
Searching master indexes for filing information ...
R> info
      cik      company.name form.type date.file quarter
1 101829 UNITED TECHNOLOGIES CORP DE 10-K 2005-02-10      1
2 101829 UNITED TECHNOLOGIES CORP DE DEF 14A 2005-02-25      1
3 101829 UNITED TECHNOLOGIES CORP DE 10-K 2006-02-09      1
4 101829 UNITED TECHNOLOGIES CORP DE DEF 14A 2006-03-09      1
```

The yearly master index files generated using `getMasterIndex` function contain filing information along with partial links for the complete filings uploaded on the SEC's EDGAR server. `getFilings` function facilitates downloading of filings by taking CIK(s), form type(s), filing year(s), and filing quarter(s) as function parameters. The following is an example for implementing this function.

```
R> output <- getFilings(cik.no = c(1000180, 38079), c('10-K','DEF 14A'),
+   2006, quarter = c(1, 2, 3), downl.permit = "h",
+   useragent)
Total number of filings to be downloaded = 4.
Do you want to download (y/n)? y
Downloading filings. Please wait...
|=====| 100%
```

The `getFilings` function downloads complete submission filings, which are in text format, from the SEC server. A user may want to take a look at these filings in HTML format. The `getFilingsHTML` function takes CIK(s), form type(s), filing year(s), and quarter(s) of the filing as user inputs. It then reads the downloaded filing, scraps the filing excluding exhibits, and saves the filing content in HTML format in the directory "*Edgar filings_HTML view*".

3.4. Extract filing header information and search filings for input keywords

Analysts may need filing header information for a firm, such as the period of the report, SIC code, business address. The `getFilingHeader` function takes an input of CIK(s), form type(s), and filing year(s), and scrapes header information of the required filings. The following code illustrates its usage.

```
R> header.df <- getFilingHeader(cik.no = c('1000180', '38079'),
+   form.type = '10-K', filing.year = 2006,
+   useragent)
```

Researchers often use qualitative information on financial reports. Especially a plethora of studies use count of specific keywords mentioned in financial reports to develop a qualitative proxy. `edgar` package provides a `searchFilings` function that searches filings for a user keyword list and returns the count of its mentions (`nword.hits`) along with filing information. A user needs to provide a search keyword list along with CIK(s), form type(s), and filing year(s). The following code demonstrates the use of this function.

⁴ By default, this function provides information on all form types filed in all the quarters of the input year(s).

```
R> word.list <- c('foreign exchange exposures','currency transactions')
R> output <- searchFilings(cik.no = c('1000180', '38079'),
+   form.type = c('10-K', '10-K405','10KSB', '10KSB40'),
+   filing.year = c(2005, 2006), word.list, useragent)
R> output
      cik      company.name form.type date.file nword.hits
1 1000180 SANDISK CORP 10-K 2005-03-18      3
2 1000180 SANDISK CORP 10-K 2006-03-15      5
3 38079 FOREST OIL CORP 10-K 2005-03-15      0
4 38079 FOREST OIL CORP 10-K 2006-03-16      0
```

The `searchFilings` function also generates detailed search result for each filing in the directory "*Keyword search results*", in HTML format. With this search results, a user can see exact position of the input words in the filing and other surrounding text of at most 250 characters. For example, the generated file "*Keyword search results->1000180_10-K_2005-03-18_0000950134-05-005462.html*" from the previous command shows the following search result.

CIK: 1000180
Company Name: SANDISK CORP
Form Type: 10-K
Filing Date: 2005-03-18
Accession Number: 0000950134-05-005462

Keywords search: 'foreign exchange exposures','currency transactions'
Number of word hits: 3

Detailed search result

..... ry from our Toshiba venture and our investments in those ventures are denominated in Japanese yen. Additionally we expect over time to increase the percentage of our sales denominated in currencies other than the United States dollar. Management of these **foreign exchange exposures** and the hedging mechanisms used to mitigate those exposures is complicated and we have limited experience in these activities. If we do not successfully manage our **foreign exchange exposures** our business results of operations and financial condition would

..... nited States Japan EMEA and non-Japan Asia-Pacific performs ongoing credit evaluations of its customers financial condition and generally requires no collateral. Off Balance Sheet Risk. The Company has off balance sheet financial obligations. See Note 5. **foreign exchange exposures**. The Company is exposed to foreign currency exchange rate risk inherent in sales cost of sales and assets and liabilities denominated in currencies other than the United States Dollar. The Company did not hedge its foreign currency risk in 2004,2003 and

3.5. Extract business description and MD&A sections from annual statements

In recent years, research using the textual analysis of firms' product/business description section and Management's Discussion and Analysis (MD&A) section in 10-K has witnessed an exponential increase [e.g.,17–19].

The `getBusinDescr` (`getMgmtDisc`) function in `edgar` package facilitates analysts and researchers to extract business description (MD&A) information for desired firms in a single command. It uses firm CIK(s) and filing year(s) as input parameters. It sequentially reads 10-K filings, removes HTML tags, extracts business description (MD&A) sections, and stores them in text files in the directory "*Business descriptions text*" ("*MD&A section text*"). These functions also return a dataframe with filing information and the extraction status, with the value of one being successfully extracted.

```
R> output <- getBusinDescr(cik.no = c(1000180, 38079),
+   filing.year = 2005, useragent)
```

```
R> output <- getMgmtDisc(cik.no = c(1000180, 38079),
+   filing.year = c(2005, 2006), useragent)
```

3.6. Retrieve form 8-K items information

The `get8KItems` function provides a tool to extract Form 8-K events. This function takes firm CIK(s) and filing year(s) as input parameters. It downloads the required 8-K filings from SEC, reads them sequentially, and extracts event information using regular expressions. The output dataframe contains Form 8-K events information along with filing information. The following code illustrates the use of this function.

```
R> output <- get8KItems(cik.no = c(1000180,38079),
+   filing.year = c(2005, 2006), useragent)
```


3.7. Generate sentiment measures of SEC filings

The `getSentiment` function provides sentiment measures of SEC filings. It takes the help of Loughran–McDonald (LM) sentiment dictionaries [20] to compute sentiment measures of the filing text. The following code illustrates its usage.

```
R> senti.df <- getSentiment(cik.no = c('1000180', '38079'),
+                           form.type = c('10-K', '10-Q'),
+                           filing.year = 2006, useragent)
```

4. Impact

The popularity of using financial statements filed on the SEC has exponentially increased since the last decade. The usage of these corporate filings are growing in multitude of areas, such as law, accounting, finance, marketing, management, statistics, environment science. However, SEC's web server provides a single filing at a time. This creates a need of automating download of these filings in bulk with an ease. Additionally, the growing topics using text-mining of SEC filings call attention to develop a tool that helps analysts and researchers for preprocessing of these filings. To fulfill this gap, `edgar` R package provides functions for downloading, parsing, searching, and sentiment analysis of filings.

5. Conclusions

Post-2000 era has seen an unprecedented rise in the textual analyses research, using financial and operational disclosure of firms, leading to an increased demand for an efficient open-source platform to download and analyze the disclosures. This paper introduces `edgar` R package in details and could serve as a primer for researchers, practitioners, and investors alike to achieve their respective goals using SEC EDGAR filings. This package works on major operating systems with greater simplicity, providing 11 functions to facilitate retrieving, storing, searching, and parsing of all the available filings on the SEC's EDGAR server.

The `edgar` package have been undergone several updations since its implantation in 2015. We plan to extend its features by adding functions to scrape other important information from filings based on the future requirements by the academicians and analysts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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